

Natural Language Processing

1. Introduction

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1. Definition and History of NLP

Natural Language and Artificial Language

❖ Natural Language

- Human language that has developed naturally and is used in everyday life (e.g., Korean, English)
- Difficult for computers to directly understand due to its flexible and unstructured form

❖ Artificial Language

- Human-designed languages such as programming languages (e.g., C, Python, Java)
- Clearly defined structure enables computers (compilers) to directly interpret and process them

Related Subjects

❖ Natural Language Processing (NLP)

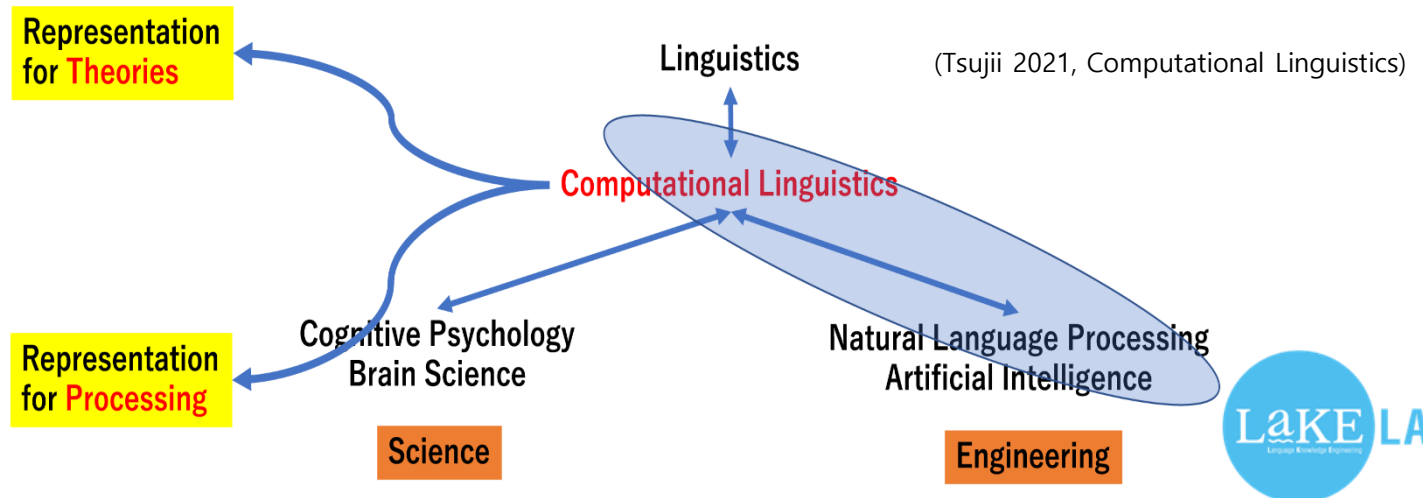
- Analyzes and utilizes natural language from a computational perspective
- Examples: morphological analysis, syntactic parsing, document proofreading, machine translation

❖ Linguistics

- Studies how human model and represent the world through natural language

❖ Computational Linguistics

- A branch of linguistics
- Focuses on how humans process and model language, often using computational methods



NLP history - Early

❖ Early Developments in Machine Translation

- The idea of using computers for language translation
 - Proposed by W. Weaver and A. D. Booth
- Russian–English Machine Translation
 - 1954: Georgetown University and IBM joint project
 - Word-level substitution → Direct Machine Translation

❖ Machine Translation in the 1960s

- Rapid growth of machine translation research
- Phrase-structure-based translation
 - Morphological analysis and syntactic parsing
 - Focus on identifying grammatical roles of words within sentences
 - Based on Chomsky's Phrase Structure Grammar (PSG)

NLP history – Dark Period

❖ The ALPAC Report

- Concluded that practical machine translation was not feasible at the time
- Led to a “dark period” in MT research
 - Late 1960s to early 1970s

❖ Major Issues Identified

- Increasing grammatical rules → Explosion of syntactic ambiguity
- Necessity of semantic understanding

❖ Syntactic Ambiguity Examples

- They are **flying** planes.
 - “flying” → verb (progressive) or adjective
- Time **flies like** an arrow.
 - “flies” → verb or noun
 - “like” → preposition or verb
- He saw a woman **in the garden with a telescope**.
 - Ambiguity in prepositional phrase attachment

NLP history – Growth Period

❖ Case Grammar

- Proposed by C. Fillmore (1968)
- Semantic analysis of case relations centered on the verb
 - Cases: Agent, Object, Instrument, etc.
- For each verb, analyze which semantic roles (cases) its noun arguments can take
- Helped reduce syntactic ambiguity
 - Actively developed especially for Japanese

❖ Example of Case Analysis

He opened the door with a key.

A key opened the door.

→ Both sentences yield the same underlying case structure:

Agent: *he*

Object: *door*

Instrument: *key*

NLP history – Growth Period

❖ Artificial Intelligence and Language Understanding

- 1966: ELIZA dialogue system
- Recognition of the need to store human knowledge
 - World knowledge helps resolve syntactic ambiguity
- Beginning of research on language understanding

❖ Functional Unification Grammar

- Proposed by M. Kay (1984)
- If the features of two adjacent words are compatible, they are unified into a single constituent
- Repeated unification → formation of a complete structure

❖ Related Grammar Frameworks

- **LFG (Lexical Functional Grammar)** – Bresnan (1982)
 - Focuses on functional relations between lexical items
- **GPSG (Generalized Phrase Structure Grammar)** – Gazdar (1985)
- **HPSG (Head-driven Phrase Structure Grammar)** – Pollard (1985)

NLP history – Present

❖ Statistical Machine Translation (SMT)

- 1985: Peter Brown, IBM Watson Research Center
 - Development of statistical translation models
 - Introduction of word alignment models
- Utilizes large-scale English–French parallel corpora to estimate translation probabilities
- Quickly outperformed rule-based MT systems
- Demonstrated that automatic learning is more effective than manually crafted rules

❖ Neural Machine Translation (NMT)

- Inspired by the functioning of the human brain
- RNN-based machine translation
- Attention-based neural translation (Bahdanau et al., 2015)
- Much more effective than SMT

NLP history – Present

❖ Word Vectors

- Words represented as vectors in continuous space
- Enable semantic and syntactic operations (e.g., similarity, analogy)

❖ Large Language Models (LLMs)

- Transformer architecture (Vaswani et al., 2017)
 - Self-attention mechanism
- Encoder–decoder structures built from stacked Transformers
- Pretrained on massive raw corpora
- Use optimized tokenization
 - Words segmented into efficient tokens
- Automatically learn linguistic rules and meaning from statistical relationships
- Representative models:
 - BERT
 - GPT
 - ChatGPT
 - Bard (Gemini)

2. Limitations of Linguistic Rule-Based Approaches

Ambiguity

❖ Ambiguity in Language

- Ambiguity exists at all levels: lexical, phrasal, and semantic

❖ Lexical Ambiguity (Word Sense Ambiguity)

- **bank**
 - financial institution / side of a river
- **bat**
 - flying mammal / baseball equipment
- **light**
 - Illumination / not heavy

❖ Sentence Ambiguity (Structural Ambiguity)

- **The stolen painting was found by the tree.**
 - “by the tree” → location?
 - or the tree as the agent? (unlikely but grammatically possible)
- **Beautiful flowers and soldiers**
 - Are both beautiful?
 - Or only the flowers?

Exceptions to Linguistic Rules

❖ Regularities in Language

- Words and sentences exhibit structural patterns
- Smaller grammatical units combine systematically
- Repeated patterns often imply similar meanings

❖ Exceptions to Linguistic Rules

▪ 1. Plural Forms

- Rule: Add “-s” or “-es”
 - book → books
 - box → boxes
- Exceptions:
 - child → children
 - mouse → mice
 - fish → fish

▪ 2. Past Tense Forms

- Rule: Add “-ed”
 - walk → walked
 - play → played
- Exceptions:
 - eat → ate
 - run → ran
 - go → went

Meaning Variation

❖ **Meaning Variation Beyond Grammar**

- The same grammatical structure can convey different meanings depending on context

❖ **Idiom vs. literal meaning:**

- hit the sack → go to sleep
- hit the bag → strike a bag

❖ **Contextual interpretation:**

- “My stomach hurts.” (physical pain)
- “That decision really hurts.” (emotional impact)

❖ “All grammars leak” – Edward Sapir

Linguistic Flexibility

❖ Neologisms and Semantic Change

- Words emerge, disappear, and shift in meaning over time

❖ Neologisms (newly coined words):

- selfie
- metaverse
- COVID-19

❖ Semantic shift (meaning change):

- nice → originally “foolish,” now “kind/pleasant”
- gay → originally “joyful,” later “homosexual”

Linguistic Flexibility

❖ Context-Dependent Meaning

- Meaning varies depending on part of speech
 - **stun (verb):**
 - “The explosion stunned him.” → to shock or knock unconscious
 - **stunning (adjective):**
 - “She looks stunning.” → extremely beautiful

❖ Situational Meaning

- Meaning changes depending on context and speaker intention
 - “Hey, my little tiger!”
 - tiger = brave or energetic child (metaphorical)
 - “She is a rose.”
 - rose = beautiful woman (metaphor)

Challenges in Natural Language Understanding

Ungrammatical Use

Heyyy ^^
Soooo good
Not really tho
Finally met ya

Segmentation

New York Times Square:
New York + Times Square
/ New York Times + Square

Predicates

A piece of cake
Cinderella
Thorn in one's eye

Neologisms

Cringe
Binge-watch
Ghosting

World Knowledge

Mary and Susan are sisters.
Mary and Susan are mothers.

Ambiguous Named Entities

Washington (person / state / D.C.)
Amazon (company / river)
Apple (company / fruit)

Linguistic Research Paradigms

❖ Rationalism

- Language is innate.
- Goal: discover internal grammatical rules.
- Key figure: Noam Chomsky
- Basis of rule-based approaches

❖ Empiricism

- Language is learned from experience.
- Goal: analyze language using data.
- Basis of corpus-based approaches

❖ Shift in Paradigm

- Dominant: Rationalism (mid-20th century)
- Since the 1990s: Rise of Empiricism
- Enabled by large corpora and hardware advances

3. Lecture Topics

Lecture Topics

❖ NLP Pipeline

- NL input -> Conceptual understanding -> Conceptual response -> NL output

❖ Processing Units:

- Words -> Sentence -> Paragraph -> Document

Lecture Topics

❖ Morphological Analysis

- Segmenting a given word (token) into the smallest meaningful units (morphemes) (often including part-of-speech information)

❖ Example (English):

- **unlockable**
 - un + lock + able
(prefix + verb + suffix)
- **cats**
 - cat + s
(noun + plural marker)

Lecture Topics

❖ Part-of-Speech (POS) Tagging

- Selecting the correct morpheme and assigning the appropriate part of speech in context.

❖ Example:

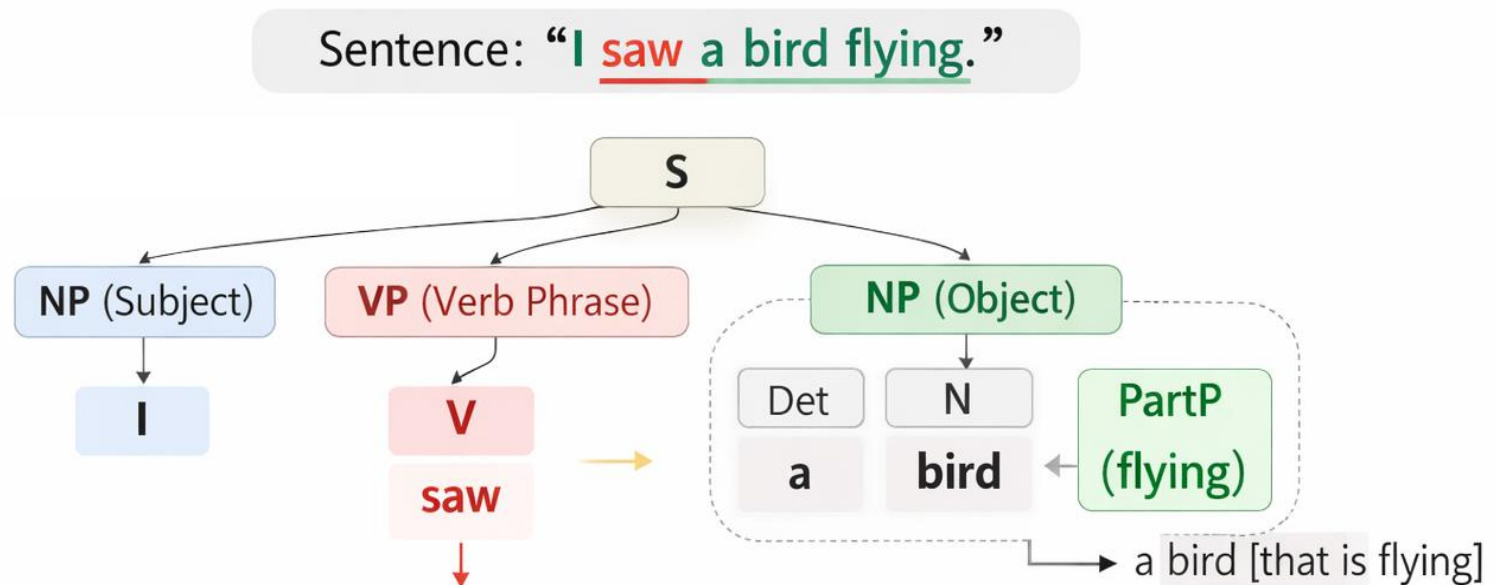
- **Flying planes can be dangerous.**
 - flying (verb / adjective)
- **I saw her duck.**
 - duck (noun / verb)

❖ POS tagging determines the correct interpretation based on the sentence context.

Lecture Topics

❖ Syntactic Parsing

- Identifying grammatical roles such as subject and object, and determining modification and dependency relations in a sentence.
- Clarifying the agent of an action



PartP: Participle Phrase

Lecture Topics

❖ Word Sense Disambiguation (WSD)

- Resolving lexical ambiguity in context.
- Determining the correct meaning of homonyms
→ different meanings within the same part of speech
- Cf: Morphological / POS tagging
→ resolves morpheme and part-of-speech ambiguity

❖ Examples:

- **I bought a *bat*.** (animal or baseball bat?)
- **She went to the *bank*.** (river bank or financial institution?)

Lecture Topics

❖ Semantic Role Labeling (SRL)

- Assigning semantic roles such as agent, patient, and experiencer.
- The syntactic subject is not always the agent.
- Semantically equivalent relations are treated uniformly.

❖ Examples:

- **He was deceived by her.**
 - Agent: *she(her)*
 - Victim: *he(him)*
- **She deceived him.**
 - Agent: *she(her)*
 - Victim: *he(him)*
- Both sentences describe the same event structure, centered on the meaning of *deceive*, with roles explicitly identified.

Lecture Topics

❖ Named Entity Recognition (NER)

- Identifying and classifying named entities such as persons, locations, and organizations as single units.
- Recognizing long or newly emerging names to improve NLP performance.

❖ Examples:

- **Elon Musk** (Person)
- **Chungbuk National University Student Council** (Organization)
- **She's Gone** (Title)
- **Catch Me If You Can** (Title)
- **12,000** (Number / Quantity)

Lecture Topics

❖ Information Extraction (IE)

- Extracting relations expressed in a sentence (using results from previous analysis stages).

❖ Examples:

- **Google acquired YouTube.**
→ Acquire(Google, YouTube)
- **YouTube was acquired by Google.**
→ Acquire(Google, YouTube)
- **Google bought YouTube for \$1.65 billion.**
→ Acquire(Google, YouTube)

❖ Different surface forms can represent the same underlying relation.

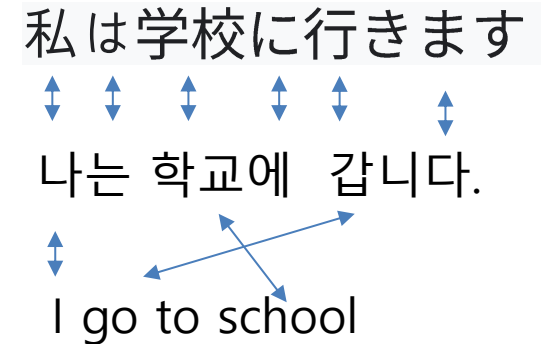
Lecture Topics

❖ Machine Translation (MT)

- Translating a source sentence into a target sentence based on linguistic analysis.

1. Word-level translation

- Lexical mapping without deep structural analysis.
- Example: Japanese to Korean (literal word mapping)



2. Syntactic-level translation

- Requires parsing to determine grammatical relations.
- Proper prepositions and word order require syntactic analysis.
- Example: Korean to English

❖ 3. Semantic-level translation

- Requires understanding intended meaning beyond literal words.
- **He kicked the bucket.**
→ *He died.*
- Literal translation fails; idiomatic meaning must be interpreted.

Lecture Topics

❖ Word Embeddings

- Encoding words as dense vectors using deep learning.
- Enable semantic vector arithmetic
king – man + woman $\approx$ queen
- Capture semantic and syntactic relationships
- Learn from large corpora without explicit dictionaries
- Distributed representations (no manual feature engineering)

Lecture Topics

❖ n-gram Language Model

- Predicting the next word based on the previous n words.
- Example (trigram):
“I would like to *drink a* ____” $\rightarrow$ *coffee*
- Larger n captures more context
- Implemented using statistical methods
- Limited by data sparsity and scalability

Lecture Topics

❖ What Is an LLM?

- **A Large Language Model (LLM)**
 - = A system that learns to predict the next word.
- Example:
 - “I drink coffee every ____”
→ morning
- By predicting the next word billions of times, the model gradually learns language patterns.
- It does not memorize rules.
It learns from data.

Lecture Topics

❖ How Does LLM Learn?

1. Reads massive amounts of text
2. Guesses the next word
3. Checks the correct answer
4. Adjusts itself slightly
5. Repeats billions of times

❖ Through this process, it learns:

- Grammar
- Word meanings
- Context
- Writing style

❖ No step-by-step linguistic pipeline is required.

Lecture Topics

❖ Why Is LLM Powerful?

❖ Because predicting the next word can become:

- Writing sentences
- Answering questions
- Translating languages
- Summarizing text

❖ But remember:

- It predicts patterns
- It does not truly “understand”
- It can generate incorrect answers